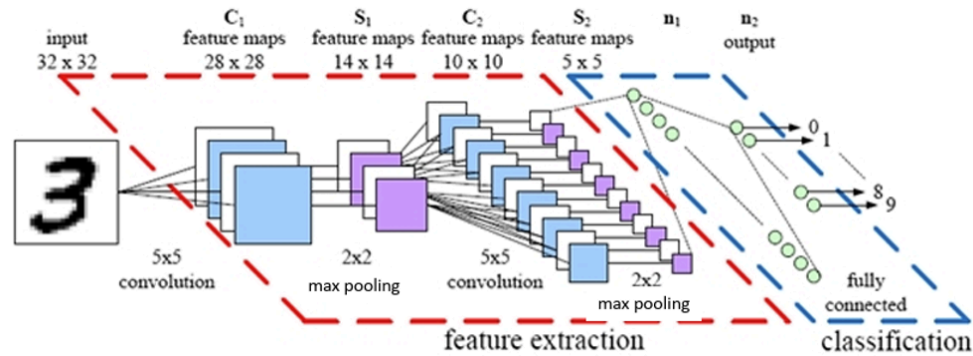


Multi-class Classification

The theory and architectures we have studied until now can be extended to handle multi-class classification problems using **Softmax Classification**.



1. Softmax (Multinomial Logistic Regression)

This method generalizes binary classification to handle multiple classes directly.

Steps:

1. For K classes, compute K linear combinations of the feature map:

$$z_k = \mathbf{w}_k^T \mathbf{x} \quad \text{for } k = 1, 2, \dots, K$$

where $\mathbf{w}_k$ contains the weights and bias for class k .

2. Apply the **softmax function** to compute the probabilities:

$$P(y = k \mid \mathbf{x}) = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}}$$

3. The predicted class is the one with the highest probability:

$$\hat{y} = \arg \max_k P(y = k \mid \mathbf{x})$$

Advantages:

- Naturally handles multi-class classification as a single model.
- Produces probabilities that are consistent and normalized (sum to 1).

Disadvantages:

- Requires solving a more complex optimization problem than binary classification.

- Training can be slower for large datasets and many classes.

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2. One-vs-Rest (OvR)

- This method is a way to expand *logistic regresison* to more than two classes.
- It involves splitting the multi-class problem into multiple binary classification problems. Each binary classifier predicts whether a sample belongs to one specific class versus all other classes.

Steps:

1. For K classes, train K logistic regression models.
2. Each model predicts the probability of a sample belonging to one class versus the rest.
3. During prediction, the class with the highest probability is selected as the final prediction.

Advantages:

- Simple to implement.
- Compatible with the binary logistic regression implementation.

Disadvantages:

- Requires training K separate models, which can be computationally expensive.
- Can sometimes lead to inconsistent probabilities across classes.

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